Lotka proposed the equations in 1910, as a way of understanding the rates of autocatalytic chemical reactions—chemical processes that regulate themselves. In the following decade, he applied the equations to the population dynamics of wild animals.

In 1926, Vito Volterra arrived at the same conclusions. He had become interested in the subject after meeting Italian marine biologist Umberto D'Ancona. D'Ancona told Volterra how the percentage of predatory fish caught in nets in the Adriatic Sea had greatly increased during World War I. This change was clearly linked to the drastic reduction in fishing during the war years, but D'Ancona could not explain why less fishing did not produce more fish of all kinds in the nets. Using the same equations as Lotka, Volterra eventually explained the fluctuations in both the predator and the prey species.